

Hayanthika Selvam

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Education

Electronics and Communication Engineering, Panimalar Institute of Technology CGPA: 8.65	2021 – 2025 Chennai, India
Higher Secondary, LFC Girls Higher Secondary School Percentage: 86.6	2020 – 2021 Ranipet, India
Secondary, LFC Girls Higher Secondary School Percentage: 79.4	2018 – 2019 Ranipet, India

Skills

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|----------------------------|----------------|
| • Python | • SQL |
| • Machine Learning | • Data Analyst |
| • Internet of Things (IoT) | • Embedded C |

Projects

ATM Security and Surveillance System Using IoT

- Developed an IoT-based security system using ESP32 for centralized control.
- Integrated RFID module for secure door access and servo motor for automatic locking.
- Implemented vibration detection to trigger a buzzer alarm and lock the door.
- Used GSM module to send SMS alerts upon unauthorized access detection.
- Incorporated ESP32-CAM for real-time surveillance with remote monitoring.
- Designed a user-friendly system with multi-layered security (RFID + sensors + alerts)

Smart Street Light

- Developed a mini project to control street lights using embedded systems for enhanced energy efficiency.
- Implemented automatic dimming features based on ambient light conditions to reduce energy consumption.
- Integrated motion sensors to activate lights when pedestrians or vehicles are detected, improving safety.
- Enabled remote monitoring and control capabilities for real-time management and maintenance.

Text to Image Conversion

- Created a mini project in Python and AIML that transforms textual descriptions into images.
- Employed advanced machine learning algorithms to improve the precision of image generation.
- Integrated natural language processing methods to accurately interpret and process user inputs.

Publications

ATM Security and Surveillance System Using IoT Server,

International Journal of Scientific Development and Research (IJSDR)

Paper ID:IJSDR2505125. Published: 2025

Designed and implemented an IoT-based security system for ATMs featuring vibration detection, servo-controlled locking, live video streaming with ESP32-CAM, RFID-based access control, and real-time alerts.

Demonstrated practical applications of embedded systems and IoT in enhancing ATM safety and surveillance.

Contributed to advancements in low-cost, scalable security solutions for financial institutions.

Paper Presentation

Machine Learning Algorithm for Driver Rating

Presented a paper on machine learning algorithms for driver rating at the 6th IConIC conference held on April 28-29, 2023.

Workshop

UI/UX Design

Attended a workshop on UI/UX.

Machine Learning

Attended a two day workshop on Machine Learning

Robotics

Participated in a two day workshop of robotics.

Internship

Internship in IIoT with Tamil Nadu Smart and Advanced Manufacturing Center Completed an internship focusing on IIoT use cases, utilizing Siemens Insights Hub for 24/7 machine monitoring and control. Developed a dashboard that connects microcontrollers to Node-RED and Insights Hub using MQTT protocol. Implemented anomaly detection and forecasting techniques to identify abnormal conditions and potential failures.

Certification

NASSCOM: Cybersecurity Analysis and Designs

I learned Cyber Security Analysis and Design from NASSCOM, which focuses on identifying security risks and protecting digital systems. This course helped me understand how to analyze vulnerabilities in networks and computer systems. It also introduced the design of secure systems to prevent cyber threats and attacks.

Language

Tamil

English